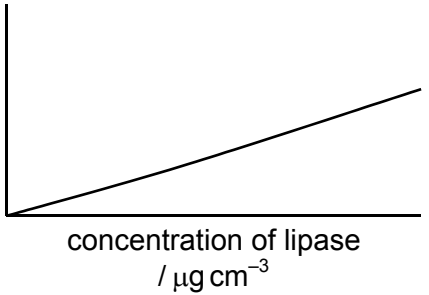


1(a)	$\left\{ \begin{array}{l} 85 \text{ (mg cm}^{-3}\text{)} \\ \frac{85 \text{ (mg cm}^{-3}\text{)} \times 35 \text{ (cm}^3\text{)}}{10\,000 \text{ (cm}^3\text{)}} \end{array} \right\}$ = 0.30 (mg cm⁻³) ;	1	A 0.298 / 0.297
1(b)	ref. to a named method of dilution or a description in words or a diagram ; ref. to using correct proportions of stock solution and water to make 100 cm³ ; 700 µg cm⁻³ and 200 µg cm⁻³ plus sensible choice of at least three other intermediate concentrations described ; e.g. (700) 575, 450, 325 (200) µg cm⁻³ (700) 600, 500, 400 300 (200) µg cm⁻³	3	A all marks on a fully labelled table
1(c)	<i>independent:</i> enzyme / lipase, <u>concentration</u> ; <i>dependent:</i> time (for pH) to reach 6.5 ;	2	
1(d)	<i>any six from:</i> 1 ref. to using same volume of each concentration of lipase (in boiling tube) ; 2 ref. to method of measuring volume ; 3 add, acid / alkali, to give pH 7.9 ;	7	2 e.g. graduated pipette / measuring cylinder / syringe / burette 3 R if buffer used

Question	Answer	Marks	Guidance
1(d)	4 <i>idea of</i> incubating enzyme (so it reaches the test temperature) before adding cotton fabric ; 5 ref. to suitable incubation temperature ; 6 ref. to timing until <u>pH reaches 6.5</u> ; 7 ref. to a suitable control ; e.g. boiled enzyme / water with alkali 8 ref. to suitable method of keeping same incubation temperature for all concentrations ; 9 ref. to using same, size / area, of stained fabric ; 10 ref. to, agitation / mixing / stirring ; 11 ref. to low / medium risk or hazard and suitable safety precaution ; 12 ref. to minimum of 3 replicates / replicating the whole experiment at least 3 times and finding a <u>mean</u> / to identify (and remove) anomalies <i>plus one required marking point:</i> 13 ref. to immersing the cloth and starting a, stop-clock / timer, simultaneously ;		5 A a single temp range 20–60 °C 8 e.g. incubator, water-bath, temperature-controlled room I air conditioning 9 A 20 mm × 20 mm pieces 11 e.g. allergy / sensitivity, to enzymes and wearing, gloves / eye protection 13 A starting, stop-clock / timer, immediately after immersing the cloth A ref. to pH meter instead of cloth since both will be immersed at the same time according to Fig. 1.1
1(e)(i)	$\frac{1}{\text{time (for pH to fall to 6.5)}} ;$	1	

Question	Answer	Marks	Guidance
1(e)(ii)	1 axes correctly orientated with labels ; x-axis: <u>concentration</u> of lipase y-axis: <u>rate</u> of, release of fatty acids / digestion of fats 2 axes have units ; x-axis unit: $\mu\text{g cm}^{-3}$ y-axis unit: s^{-1} or min^{-1} 3 line shows increase as lipase concentration increases ; e.g. rate of release of fatty acids $/\text{s}^{-1}$ or min^{-1}  concentration of lipase $/\mu\text{g cm}^{-3}$	3	2 A arbitrary units or 1 / t(ime) on y-axis 3 A linear curve or plateau
1(e)(iii)	<i>idea of rates</i> calculated from times for pH to reach 6.5 for washing liquids (from Table 1.1) ; use of intercepts / described ;	2	
1(f)	<i>any one from:</i> <i>idea that</i> the lipase in A has a higher activity than that of B ; <i>idea that</i> lipases are of, different types / different origin ; <i>idea that</i> washing liquids and test solution are different (from each other) ; e.g. washing liquids contain other chemicals to remove fat	1	I ref. to optimum temperature